



Search

Extraction of zinc

Download as PPTX, PDF • 1 like • 2,158 views **AI-enhanced description**

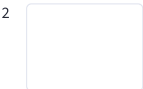
K **koshykanjirapallikaran** [Follow](#)

The document discusses the extraction of zinc from its ores. Zinc is extracted through either pyrometallurgical or hydrometallurgical extraction processe... [Read more](#)

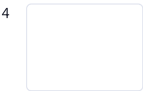
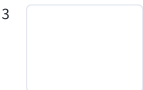
1 of 16

Download now

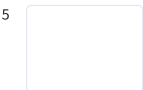
Downloaded 28 times



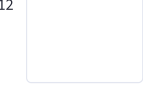
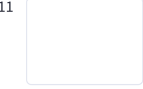
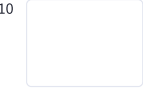
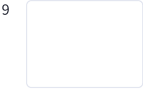
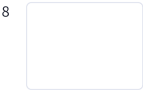
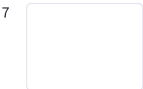
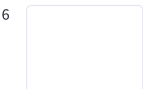
Most read



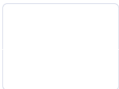
Most read



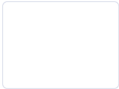
Most read



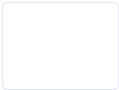
13



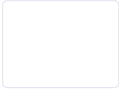
14



15



16

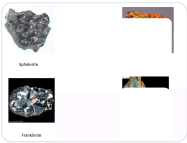


Extraction of Zn

Presented by
Nikhil
Mishra
145 09/22

Introduction

- Zinc is mainly used for
 - Galvanizing
 - Production of alloys
 - Dry cell batteries
- Its minerals are
 - Sphalerite (ZnS)
 - Zincite (ZnO)
 - Smithsonite ($ZnCO_3$)
 - Franklinite ($ZnO \cdot Fe(OH)_2$)



AD

Pyro metallurgical extraction of Zinc



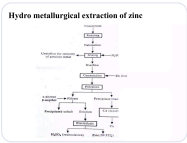
Procedures

- The ore is first ground and subjected to float
- The concentrate obtained is subjected to roast
- Roasted oxide is ground and agglomerated to a 100mm. Used entering and sent for sinter

$$\text{ZnS} + 1.5\text{O}_2 \rightarrow \text{ZnO} + \text{SO}_2 \quad (\text{re})$$
$$\text{ZnO} + \text{CO} \rightarrow \text{Zn} (\text{g}) + \text{CO}_2 \quad (\text{re})$$
$$\text{C} + \text{CO}_2 \rightarrow 2\text{CO}$$

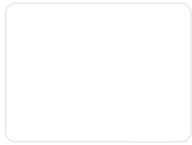
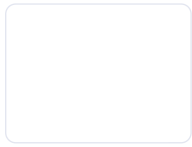
AD

- ZnO is reduced by carbon
- Make sure CO partial pressure is maintains reaction
- Zinc vapour obtained is condensed at 450°C
- Zinc condensation has to be carried out quickly so process formation of base oxide of Zn (ZnO)

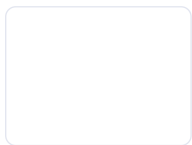
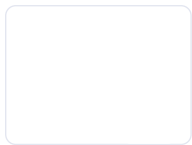


AD

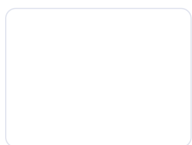
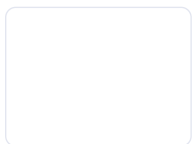
AD



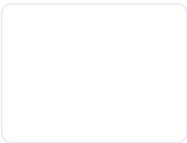
AD



AD



AD



More Related Content

What's hot (20)

- PDF

Non - Ferrous Extraction of Metals Lecture Notes

FellowBuddy.com
78 slides • 4K views
- PPTX

Faisal

Gulfam Hussain
19 slides • 7K views
- PDF

Extraction of magnesium (mg)

Vikas Barnwal
19 slides • 1.8K views
- PPTX

Zinc final

Deepansha Kakkar
40 slides • 3.8K views
- PPTX

Novel electrowinning technologies.ppt copy

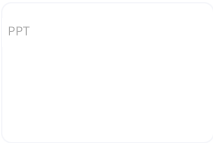
khileshkrbhandari
25 slides • 7.1K views
- PPTX

Extraction of zinc

Jyothireddy Inturi
17 slides • 369 views
- PDF

CHAPTER 7 Extraction of metals from sulphide ores.pdf

Weldebrhan Tesfaye
43 slides • 976 views



Cathodic and anodic protection

THOMAS THANGADURAI K

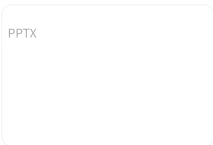
15 slides • 49.8K views



Extraction of Titanium

Vikas Barnwal

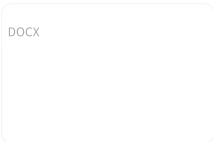
9 slides • 1.5K views



Deoxidation and inclusion control

Nikhi Lesh

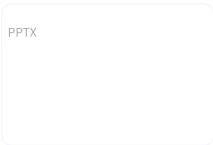
21 slides • 3.6K views



Extractive Metallurgy

AliAbbasAslam

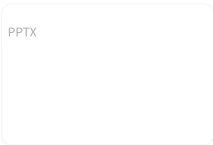
11 slides • 1.9K views



7 lead extraction

Imtiaz Ali Soomro

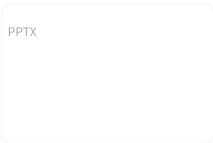
16 slides • 2.1K views



Magnesium extraction

Jyothireddy Inturi

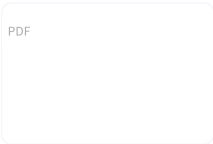
24 slides • 717 views



Steel making process

Taimoor Hassan

14 slides • 1.8K views



Electrometallurgy_week 7_8.pdf

FatimaAzharAzharIqba

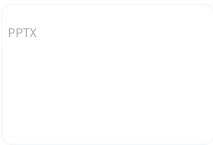
21 slides • 1.7K views



Steel Making: Lecture deoxidation

NED University of Engineering and Technology

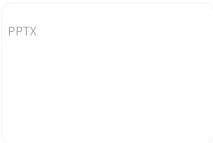
8 slides • 2.6K views



Lead Extraction

Tushar Khan

19 slides • 2.2K views



Extraction of Aluminium

SB Solutions

12 slides • 1.5K views



Introduction to steel making processes

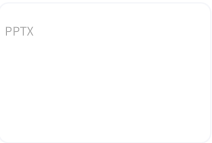
pooja verma
40 slides • 15.8K views



Extraction of Lead

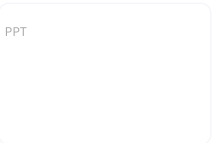
Imtiaz Ali Soomro
16 slides • 13.7K views

Viewers also liked (15)



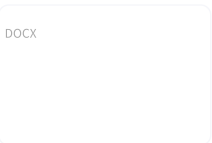
Farming systems

Saed Manasrah
11 slides • 494 views



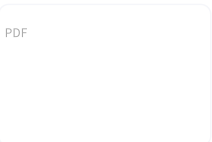
How to inform your clients

Xin Yi Zyx
7 slides • 247 views



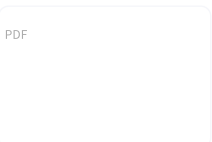
Matrks

kusnadiyoan
19 slides • 1K views



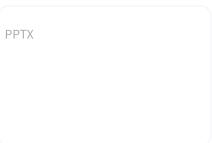
Inf financiera 2010

Claudio Venegas Bravo
44 slides • 109 views



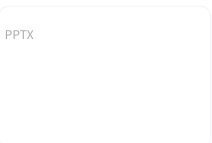
Union Elementary Principals - Deoffice, Less Office, No Office

Michael Niehoff
26 slides • 319 views



Presentation design activity

Dan Morse
11 slides • 274 views



ATreVEE IN: Using Natural Interaction in Procedure Simulator for Training in ...

Tatiana Tavares
19 slides • 282 views



Ieee 2014 2015 matlab projects titles list globalsoft technologies

IEEEMATLABPROJECTS
35 slides • 474 views



Ieee 2014 2015 matlab projects completed final list

IEEEMATLABPROJECTS

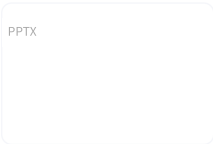
		27 slides • 347 views
PPTX	Siba 24062015 john c presentation (1) SIBA	29 slides • 261 views
PDF	John LEGERE vic54diesel	6 slides • 191 views
PPTX	Іжкписследования константина иовкова BuTCHeR1337	5 slides • 223 views
PPT	È possibile contrastare il surriscaldamento? Consorzio LaMMA - Corso UdC	10 slides • 243 views
PDF	Datini vol 1 Vasile Mariana	504 slides • 265 views
PPTX	ПремияTrainings Тренинг-Бутик HRTrainingsExpo2014	15 slides • 332 views

Similar to Extraction of zinc (20)

PPT	Extractive Metallurgy Project Abhijeet Singh	35 slides • 4K views
DOCX	Zinc Gandaki Boarding School,Lamachaur-16 Pokhara, Nepal	2 slides • 605 views
PDF	Zinc Metallurgy Pushkar Raj Chandna	5 slides • 6.7K views
PPTX	METALLURGY.pptx science class 1 to std 10 mishrakhushi7296	27 slides • 47 views
PPT	lead smelting.ppt	



Imran Shahnawaz
22 slides • 325 views



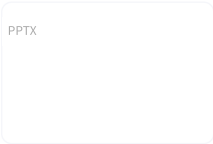
Metallurgy based on chemistry the reactions and processes

Justin George
32 slides • 5.7K views



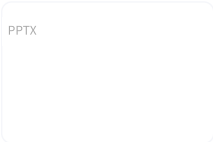
Procedure of extracting lead metallurgy.pptx

buttisoda6
6 slides • 13 views



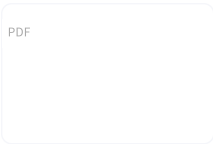
Metallurgy

RAICHEL NIVETHA X
40 slides • 274 views



metallurgy (2).pptx

WG WG
32 slides • 73 views



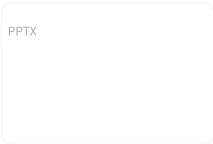
Metallurgy is a part of chemistry which deals with the extraction of metals.

nakkasatyakailash
42 slides • 135 views



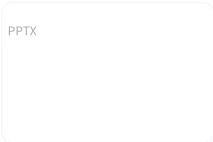
Tin and lead extraction Presentation (7).pdf

fpro7016
32 slides • 50 views



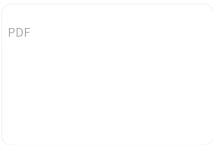
General principles of metals extraction

Mr. Muhammad Ullah Afridi
27 slides • 3.9K views



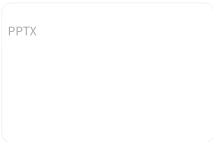
Talk 2-Hydromet-Cementation non ferrous.pptx

amitkr561
14 slides • 15 views



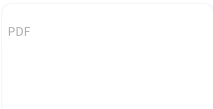
Ijciet 10 01_188

IAEME Publication
7 slides • 36 views



Sathityatiwat ChIR thesis defense

Suthapat Sathityatiwat
43 slides • 104 views



Pem ln 1_mkp

Dr. M. Kalyan Phani

	35 slides • 643 views
DOCX	Chemistry project nirmal singh 19 slides • 29.1K views
PPTX	General principles and processes of isolation of elements Mukul Kumar 35 slides • 32.3K views
PDF	generalprinciplesandprocessesofisolationofelements.pdf LUXMIKANTGIRI 35 slides • 30 views
PPTX	General principles and processes of isolation of elements niralipatil 35 slides • 1.2K views

Recently uploaded (20)

PPTX	MULTI LEVEL DATA TRACKING USING COOJA.pptx dollysharma12ab 12 slides • 130 views
PDF	4 Tier Teamcenter Installation part1.pdf VnyKumar1 2 slides • 119 views
PPTX	Water resources Engineering GIS KRT.pptx Krunal Thanki 100 slides • 118 views
PDF	20ME702-Mechatronics-UNIT-1,UNIT-2,UNIT-3,UNIT-4,UNIT-5, 2025-2026 Mohanumar S 282 slides • 100 views
PDF	2010_Book_EnvironmentalBioengineering (1).pdf EmilianoRodriguezTll 884 slides • 28 views
PPTX	Chapter_Seven_Construction_Reliability_Elective_III_Msc CM SubashKumarBhattarai 32 slides • 25 views
PDF	67243-Cooling and Heating & Calculation.pdf

[illegible]

149 slides • 21 views

PPTX

Precedence and Associativity in C prog. language

Mahendra Dheer

7 slides • 127 views

PDF

2025 Laurence Sigler - Advancing Decision Support. Content Management Ecommer...

Francisco Javier Mora Serrano

70 slides • 121 views

Extraction of zinc

- 1. **Extraction of Zinc** Presented by Nikhil Koshy Jacob ME Industrial Metallurgy 14MY33
- 2. **Introduction • Zinc** is mainly used for Galvanizing Production of alloys Dry cell batteries • Its minerals are Sphalerite (ZnS) Zincite (ZnO) Smithsonite (ZnCO3) Franklinite (ZnO(Fe,Mn)2O3)
- 3. **Sphalerite Zincite Franklinite** Smithsonite
- 4. **Pyro metallurgical extraction** of Zinc Jkkkkkkkkkkk
- 5. **Procedures • The** ore is first ground and subjected to froth flotation • The concentrate obtained is subjected to roasting • Roasted oxide is ground and agglomerated and then sintered at 1200 in a Dwight Lloyd sintering and sent for retort reduction
- 6. **• ZnO is** reduced by carbon • Make sure CO partial pressure is maintained to prevent reverse reaction • Zinc vapour obtained is condensed at 4500c to form liquid Zn • Zinc condensation has to be carried out quickly to prevent formation of blue oxide of Zn+ZnO
- 7. **Hydro metallurgical extraction** of zinc
- 8. **Procedures • Dissolution** of zinc calcine in sulphuric acid • Precipitation of Cd by adding zinc dust to the leach solution i.e cementation • Precipitation of cobalt from leach solution by adding alpha nitroso – beta naphthol • Adjust the pH of leach solution to 5.3 to hydrolyze Fe and Al and add of MnO2 to oxidize iron to trivalent state to produce a precipitate that can absorb As, Sb, and Ge • Do electrolysis of filterate solution
- 9. **Imperial Smelting Process** J,h
- 10. **Advantages of ISP** • Its possible to simultaneously smelt low grade complex mixed charges to recover both Zn and Pb • Recovery of Zn is less expensive • A wide variety of furnace sizes are available • Furnace operation is fully automated • The mechanism is highly robust
- 11. **Reactions Involved**
- 12. **Procedures • Preheated** coke and sinter is charged from the top of the furnace • The lead becomes liquid metal and flows to the bottom of the furnace collecting precious metals like gold , silver etc (lead bullion) • Slag and lead bullion at the bottom can be separated because of their density differences • The furnace gases produced flows upwards to the condensers • Zinc and lead are collected after condensation
- 14. **Procedures • The** condensers perform two functions ie cool the furnace gases, help in dissolving the condensed Zn in molten lead • The rotors produce a spray of lead bath • Zn is cooled rapidly and goes in to the solution with lead • Zinc containing lead is pumped out of from the condensers and cooled • During cooling the lead becomes saturated with zinc and excess zinc emerges from the lead and floats on the surface • Zinc is separated and the lead is fed back to the condensers
- 15. **Zinc from Lead** Slags by Slag Fuming • Slag produced in the blast furnace have considerable amount of zinc in it • The slag is fed to a slag fuming furnace • The following reactions take place • The ZnO fumes are collected in bags and sent for Zn recovery to a zinc plant

Download